LAB – 08

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Vertical Seismic Profile (VSP) Processing and Corridor Stack Generation

Introduction:

Vertical Seismic Profiling (VSP) is an advanced seismic survey method that places seismic receivers within a wellbore while positioning the seismic source at the surface. In Zero-Offset VSP (ZVSP), the source is located as close to the wellbore as possible. This configuration is instrumental in providing detailed subsurface insights, including precise time-depth relationships, seismic velocity profiles, and the calibration of well data with surface seismic surveys.

In this report, we process a ZVSP dataset using a sequence of steps in SeismicUnix (SU) scripts. The primary objectives are to extract the time-depth relationship, determine average and RMS velocities, and generate a corridor stack for seismic interpretation. Each step involves preprocessing the raw data, separating wave components, applying deconvolution, and deriving key seismic attributes. This workflow ensures accurate calibration of well data with surface seismic surveys, enabling improved reservoir characterization and exploration decision-making.
The following sections detail the methodology and findings, including relevant plots and tables derived from the ZVSP analysis.

Results and Interpretation:

First, we will use the **suswapbytes** command to adjust the dataset format to match our system's environment. Following this, we will execute the **0_2_utl_Display_ZVSP_RAW.sh** script to visualize the raw VSP data for initial inspection.

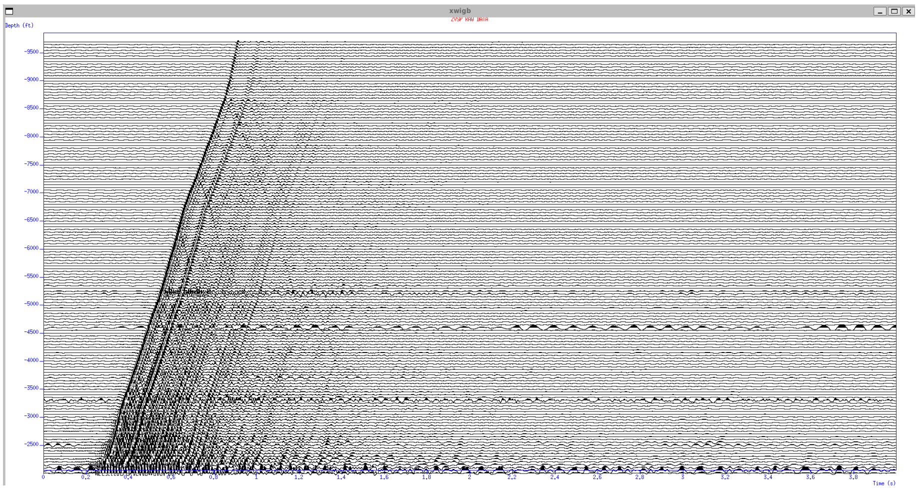


Fig. 1 RAW VSP PROFILE

Next, we will run the ***1_FBAutoPick.sh*** file to pick the first arrival. The default value of ***window_pick*** was given as 0.03, we have chosen the value as 0.015 to pick the first arrival perfectly.

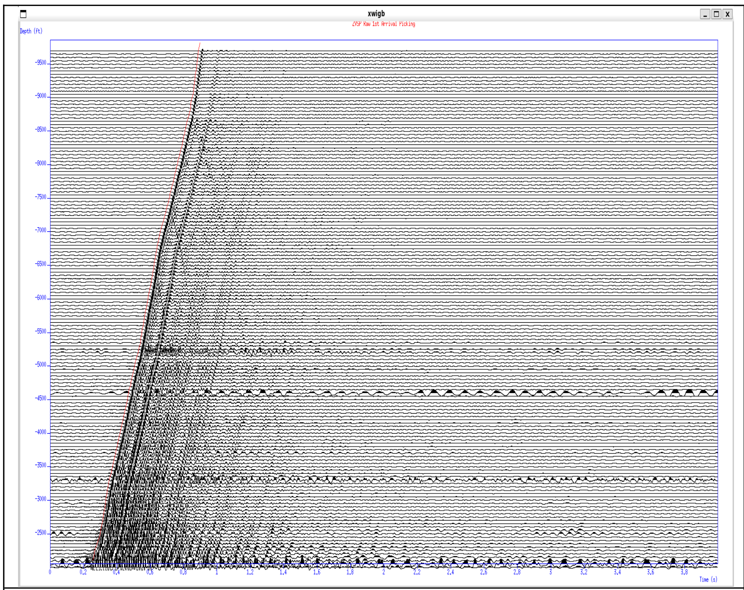


Fig. – 2 First Arrival picking using Window_size = 0.03

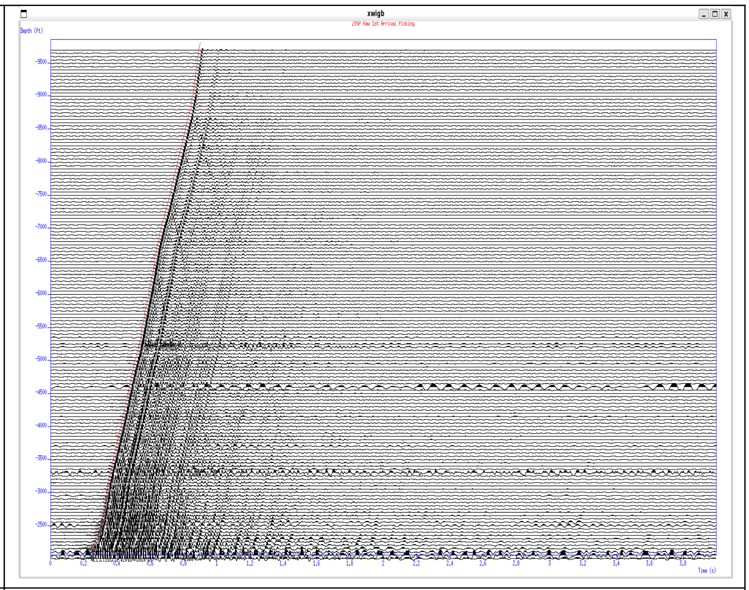


Fig. – 3 First Arrival picking using Window_size = 0.015

Next, we need to execute the `0_3_utl_CheckBPFandAmpRecovery.sh` and `0_4_utl_FrequencyAnalysis.sh` scripts to identify optimal parameters for noise reduction using F-X and F-K filtering techniques.

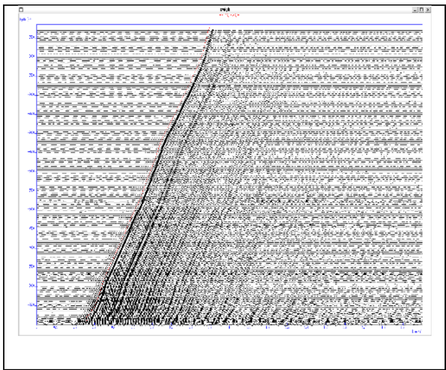


Fig. – 4 BPF = 10,15,80,95

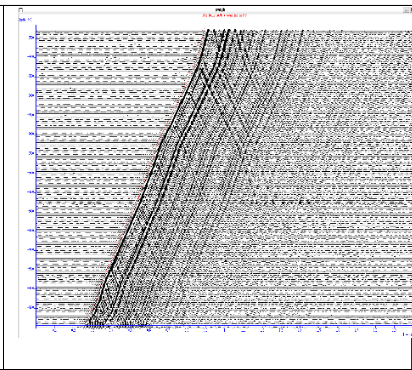


Fig. – 5 BPF + RMS Normalization

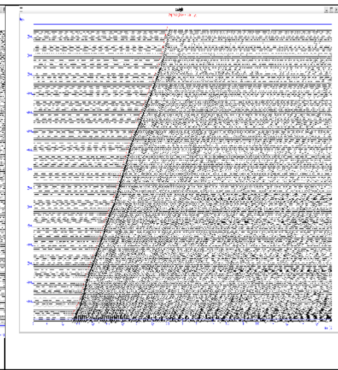


Fig – 6 BPF + tpow-1.5

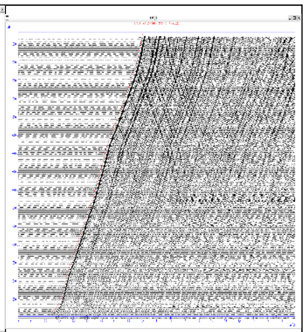


Fig. – 7 BPF + RMS + Tpow

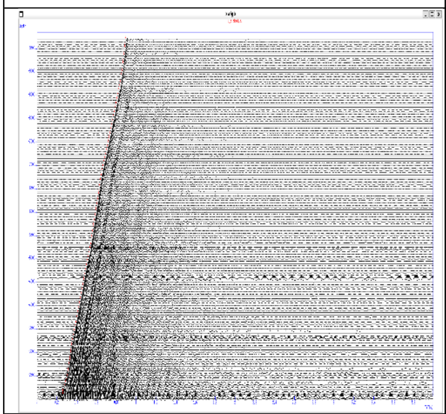


Fig. – 8 Picked Data

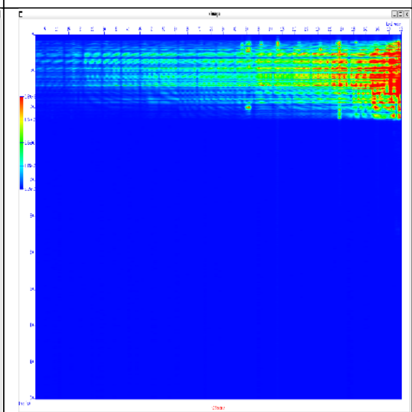


Fig. – 9 Amplitude Spectrum

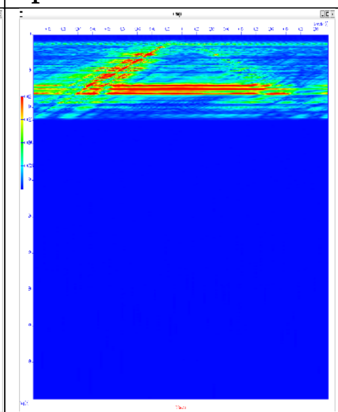


Fig. – 10 Fk Filtering

Now finally we have our desired parameters, so we can finally run the preprocessing shell script file by putting our desirable files. Finally we have our input and output files

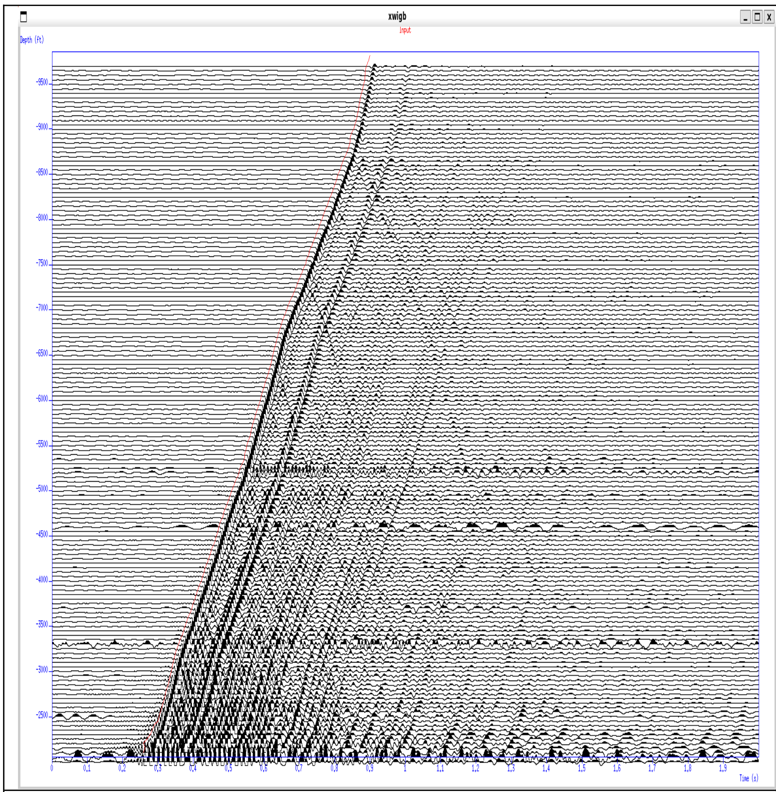


Fig. – 11 Input file data visualization

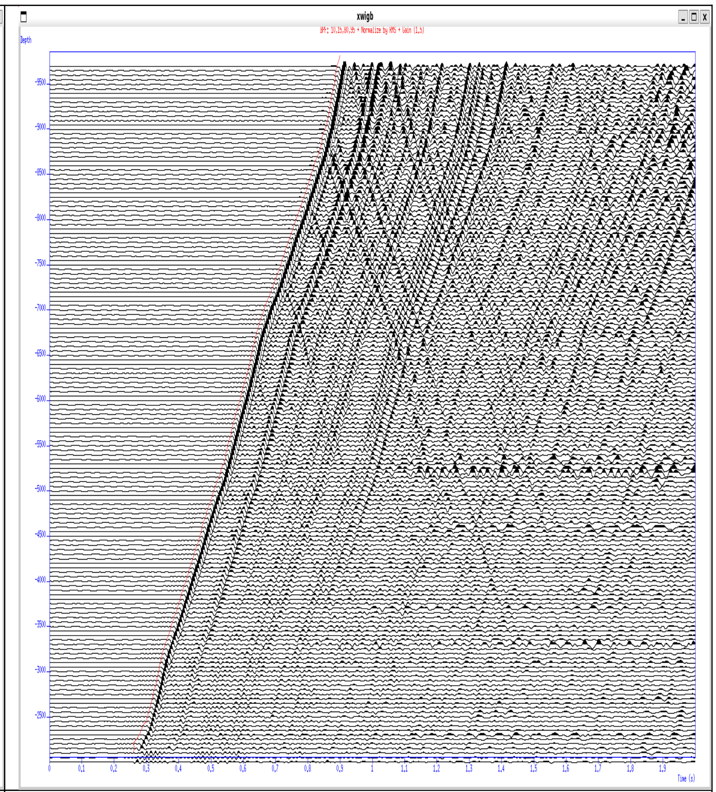


Fig. 12 – Corrected denoised and gain applied data

Now we will run ***3_Separation.sh*** file to separate the upgoing and downgoing events, and we can visualize the residual data after picking also.

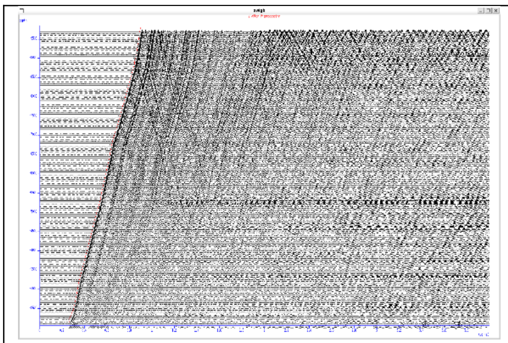


Fig. 13 – Z after Pre-processing

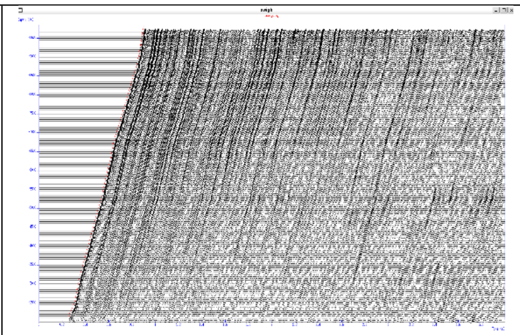


Fig. – 14 Downgoing rays

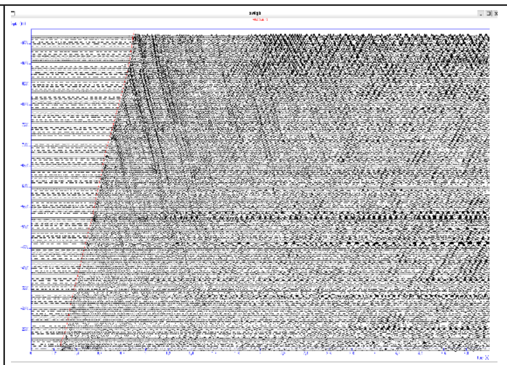


Fig. – 15 Residual - 1

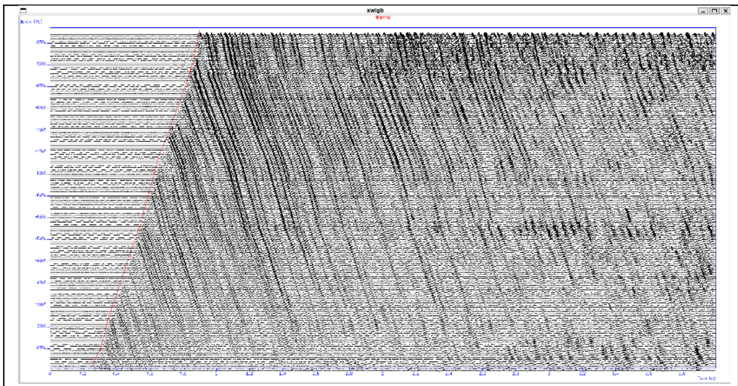


Fig. 16 – Upgoing Reflections

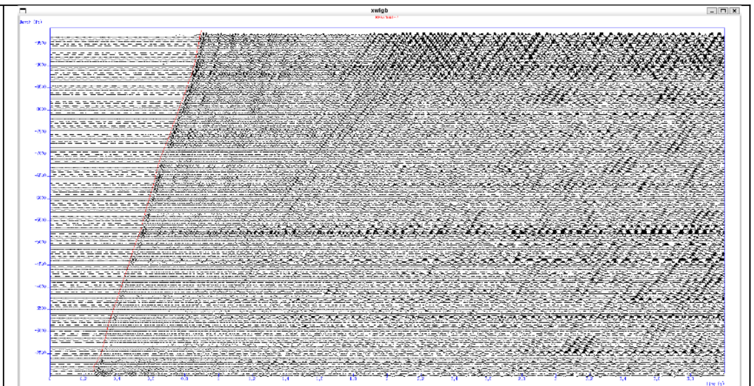
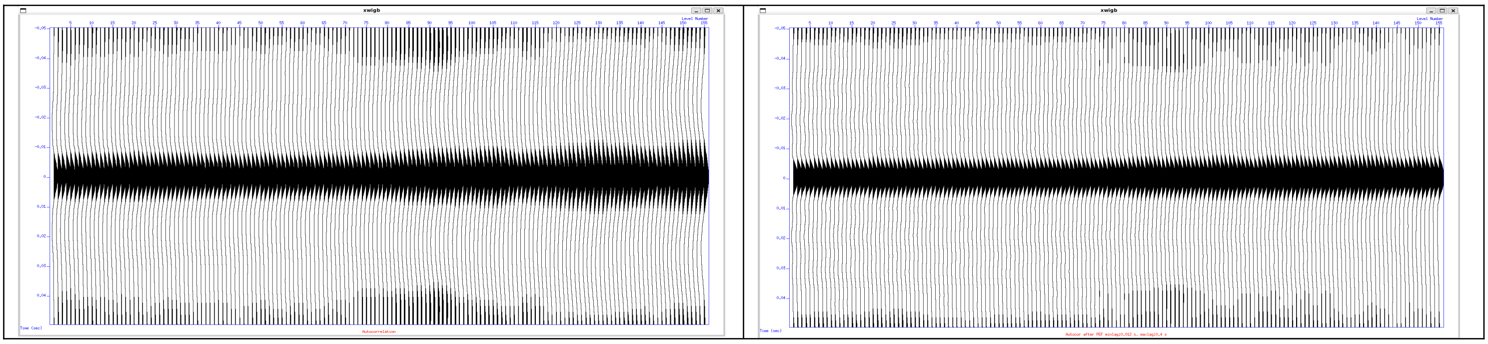


Fig. 17 – Residual 2

Next we will run the 0_5_utl_CheckAutoCorrelation.sh file to check the auto-correlation graphs. The parameters chosen are used in the deconvolutional file.



Now we will run the ***4_ApplyPEFDecon.sh*** file to get the deconvolved data. The outputs are shown below

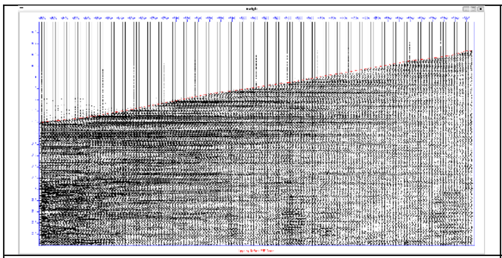


Fig. – 20 Upgoing before PEF Decon

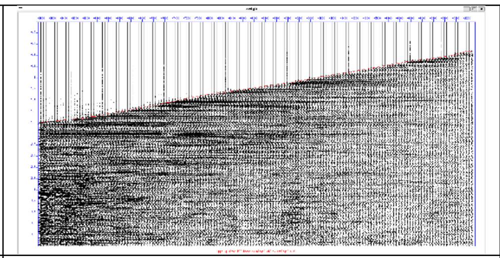


Fig. 21 Upgoing after PEF Decon

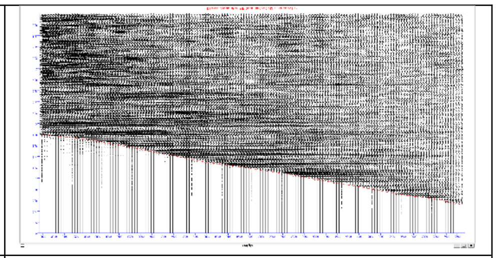


Fig. 22 Upgoing after PEF Decon(ENHANCED)

Finally we can 5_CorridorStack.sh script to get out corridor stacked image.